ResolveAI: Working Process & Architecture

1. System Overview

ResolveAI is a stateful, trust-gated Mixture-of-Experts (MoE) customer support engine. Instead of exposing a single LLM to a database (which invites hallucinations and prompt injection), ResolveAI strictly isolates reasoning from execution using a multi-node LangGraph state machine.

2. The LangGraph State Machine

Every incoming support query traverses a deterministic pipeline:

- **Intake Node:** Sanitizes the incoming request and performs initial prompt-injection screening.
- **Classifier Node:** Identifies the exact customer intent (e.g., 'refund_request', 'order_status').
- **Context Retrieval:** Fetches relevant internal CRM data (e.g., Order ORD-1001 details) and Knowledge Base articles via semantic search.
- **Resolver Node:** Generates a proposed action and calculates a mathematical confidence score.
- **Safety Gate:** Intercepts high-risk requests. If the confidence is below the intent-specific threshold (or if it's a high-value transaction like a $100 refund), execution is halted and sent to the Human-in-the-Loop console.
- **Execution Node:** Only reached if the safety gate passes. Safely executes the database mutation.

3. Mixture-of-Experts (MoE) Routing Engine

To drastically reduce API costs while maintaining high intelligence, ResolveAI triages requests to different tiers of models:

- **Tier 0 (Gemini 2.5 Flash):** Ultra-fast and cheap. Handles routine tasks like intent classification and simple FAQ responses.
- **Tier 1 (Claude 3.5 Sonnet):** Balanced cost and reasoning. Manages logistics, order tracking, and mid-level account updates.
- **Tier 2 (OpenAI o1 / Claude Opus):** Expensive frontier reasoning. Strictly reserved for high-risk queries, angry escalations, or massive refunds.

4. Cryptographic SHA-256 Audit Ledger

Absolute accountability is required for enterprise AI. Every node transition, token cost metric, and database action is logged in a tamper-evident cryptographic ledger. Each block is hashed using SHA-256 and mathematically chained to the `prev_hash` of the previous block. If any internal employee or attacker attempts to retroactively alter a log (e.g., to cover up an unauthorized refund), the hash chain breaks instantly, and the Dashboard rejects the verified badge. To ensure GDPR compliance (Right to be Forgotten), Customer PII is encrypted with a rotatable key before hashing. Shredding the key anonymizes the data without breaking the mathematical integrity of the chain.